Web Development Foundations

Week 7 – "From Working – To Clean"

Day 1: "Structure Behavior"

1

Working Is Good – Clear Is Better

Working code matters.

Clear working code matters more.

This week:

- behavior should still work
- code should be easier to read
- each function should have a clear job
- your explanation should match your structure

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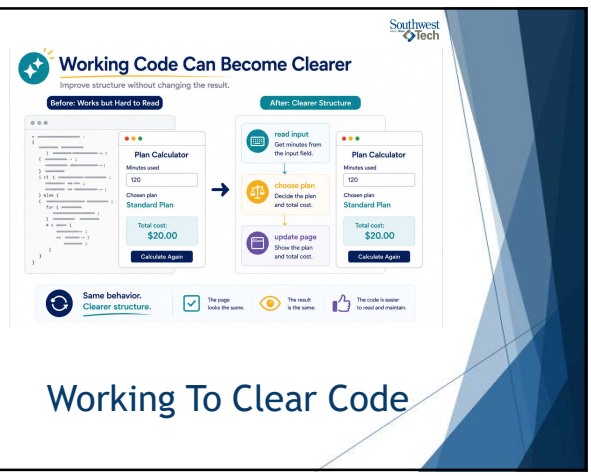

Working To Clear Code

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Lab Review: Successful Path

One successful path:

- site runs without console errors
- interaction works
- bug report explains the issue
- fix is connected to evidence
- verification proves the result

Now we will ask:

Can another person understand the code?

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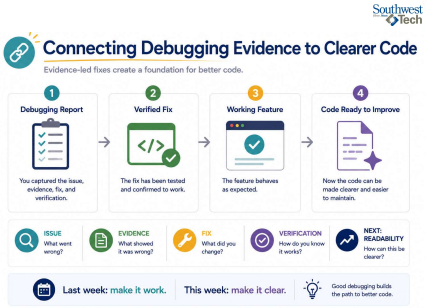

Week 6 Success Path

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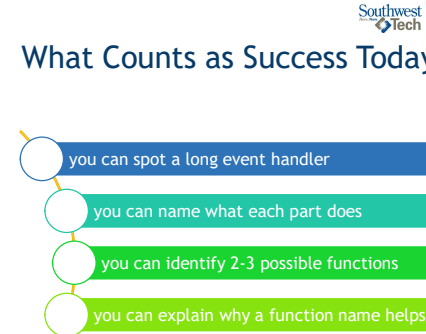

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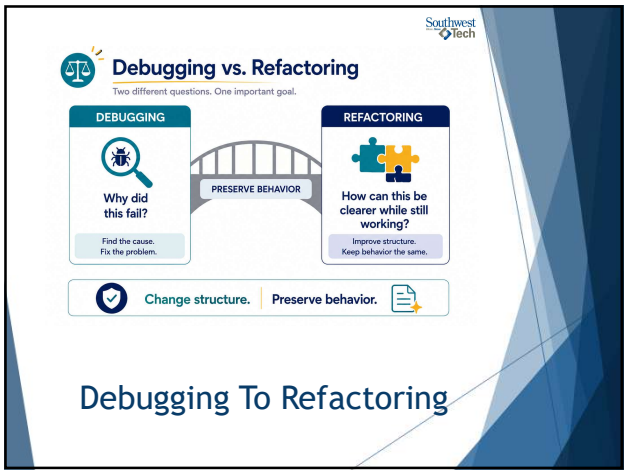
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Debugging vs. Refactoring
Two different questions. One important goal.
DEBUGGING
Why did this fail?
Find the cause. Fix the problem.
PRESERVE BEHAVIOR
REFACTORING
How can this be clearer while still working?
Improve structure. Keep behavior the same.
Change structure.
Preserve behavior.
Debugging To Refactoring

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What We Will Use Today
working behavior
function
responsibility
event handler
callback
scope
meaningful name
retest

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Today's Structured JavaScript Toolbox
Tools for building clear, organized, and reliable behavior.
working behavior
function
responsibility
event handler
callback
scope
meaningful name
retest
BUILD • TEST • IMPROVE
Use these tools together.
Write code that works, is easy to understand, and is easy to maintain.
Clear purpose
Focused responsibilities
Responds to actions
Test and confirm
Refine and improve
Structured JavaScript Toolbox

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What We Will “Skip” For Now

- modules
- classes
- complex objects
- advanced scope rules
- build tools
- framework patterns

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Parked For Later

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Code Can Work And Still Be Hard To Read

Working but messy code often has:

- one long event handler
- repeated output text
- several decisions in one place
- vague names
- hidden responsibilities

The browser may be happy.

The next human reader may not be.

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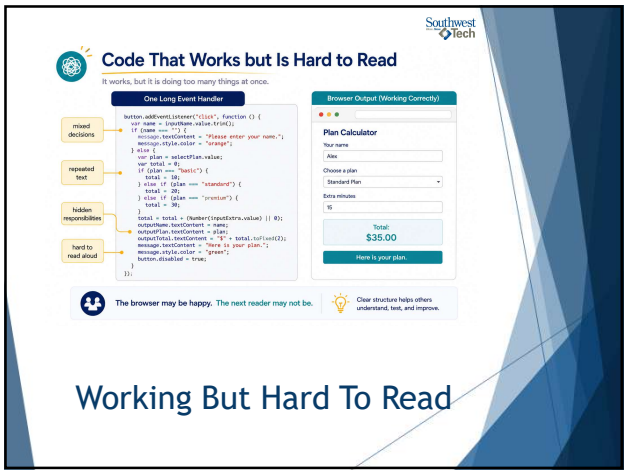
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Code That Works but Is Hard to Read
It works, but it is doing too many things at once.
One Long Event Handler
mixed decisions
repeated text
hidden responsibilities
hard to read aloud
Browser Output (Working Correctly)
Plan Calculator
Your name
Alex
Choose a plan
Standard Plan
Extra minutes
15
Total:
$35.00
Here is your plan.
The browser may be happy. The next reader may not be.
Clear structure helps others understand, test, and improve.
Working But Hard To Read

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Function As A Named Responsibility
What job does this code do?
What information does it need?
What result does it produce?
Is the name honest?
Good names make code easier to read aloud.

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Turning a Responsibility into a Named Function
Name the job. Clarify the purpose.
RESPONSIBILITY
choose the study plan
This part of the code decides which plan to use.
FUNCTION
chooseStudyPlan(minutes)
f( )
This function takes the minutes and returns the plan to use.
What job?
Choose the study plan based on the minutes.
What input?
The number of minutes to study.
What result?
The plan name (basic, standard, premium).
Good names make code easier to read.
Function As Named Responsibility

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Callback: *A Function Used Later*

In browser code, a callback is often:

- a function
- handed to another piece of code
- used later when something happens

Example:

```
`button.addEventListener("click", showPlan);`
```

The click happens later.

The function runs then.

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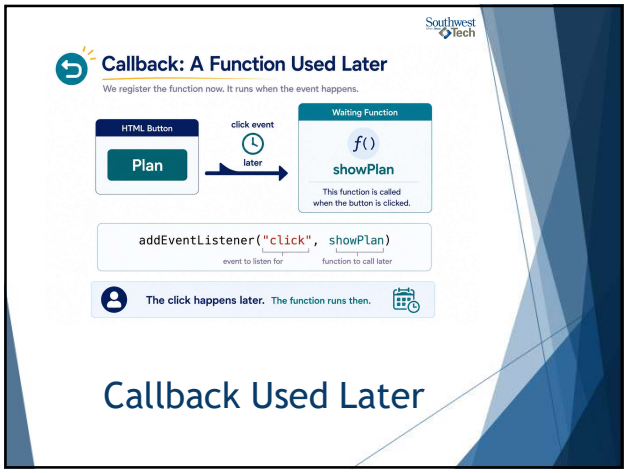

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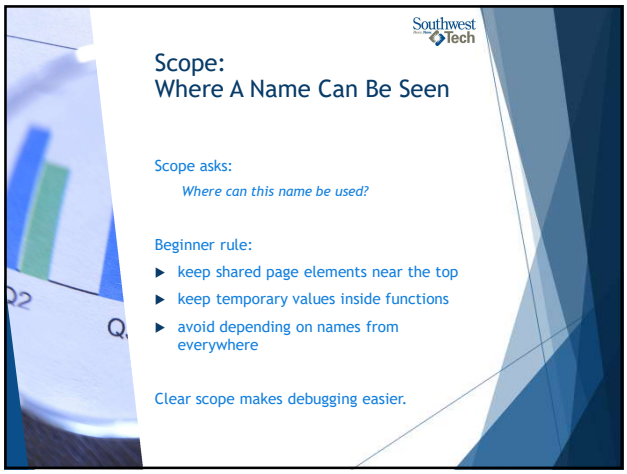

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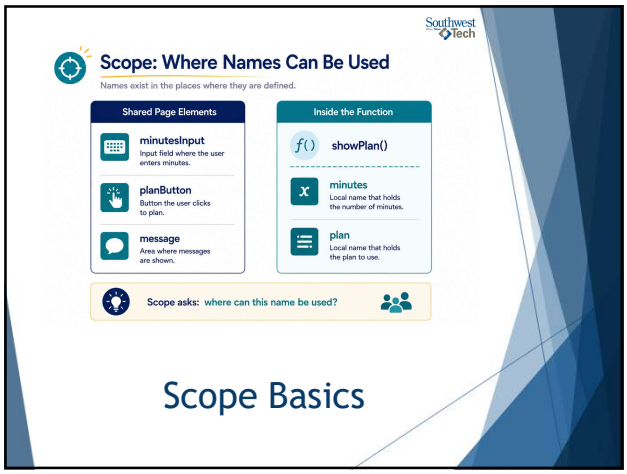
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Scope: Where Names Can Be Used
Names exist in the places where they are defined.
Shared Page Elements
minutesInput
Input field where the user enters minutes.
planButton
Button the user clicks to plan.
message
Area where messages are shown.
Inside the Function
showPlan()
minutes
Local name that holds the number of minutes.
plan
Local name that holds the plan to use.
Scope asks: where can this name be used?
Scope Basics

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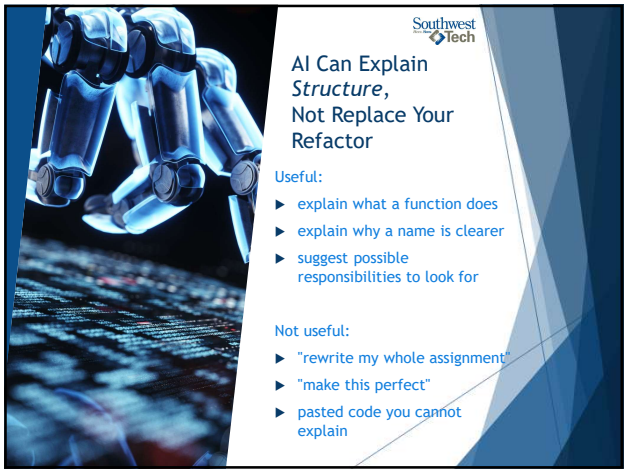
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AI Can Explain Structure, Not Replace Your Refactor
Useful:
explain what a function does
explain why a name is clearer
suggest possible responsibilities to look for
Not useful:
"rewrite my whole assignment"
"make this perfect"
pasted code you cannot explain

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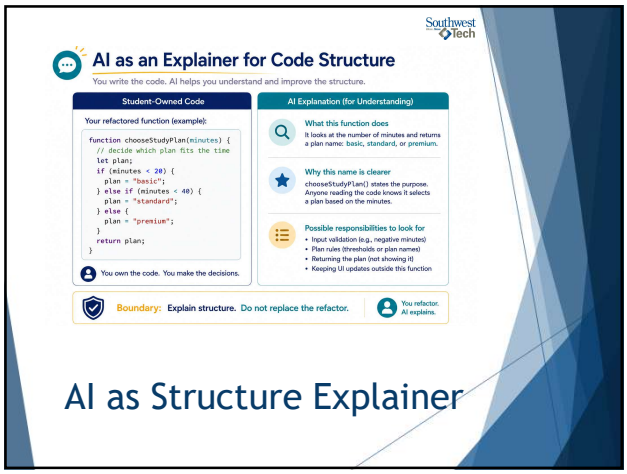
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AI as an Explainer for Code Structure
You write the code. AI helps you understand and improve the structure.
Student-Owned Code
Your refactored function (example):
function chooseStudyPlan(minutes) {
// decide which plan fits the time
let plan;
if (minutes < 20) {
plan = "basic";
} else if (minutes < 40) {
plan = "standard";
} else {
plan = "premium";
}
return plan;
}
You own the code. You make the decisions.
AI Explanation (for Understanding)
What this function does
It looks at the number of minutes and returns a plan name: basic, standard, or premium.
Why this name is clearer
chooseStudyPlan() states the purpose. Anyone reading the code knows it selects a plan based on the minutes.
Possible responsibilities to look for
Input validation (e.g., negative minutes)
Plan rules (thresholds or plan names)
Returning the plan (not showing it)
Keeping UI updates outside this function
Boundary: Explain structure. Do not replace the refactor.
You refactor. AI explains.
AI as Structure Explainer

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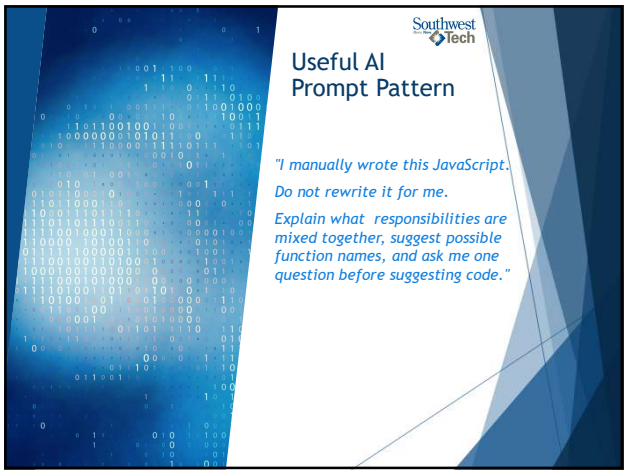
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Useful AI Prompt Pattern
"I manually wrote this JavaScript.
Do not rewrite it for me.
Explain what responsibilities are mixed together, suggest possible function names, and ask me one question before suggesting code."

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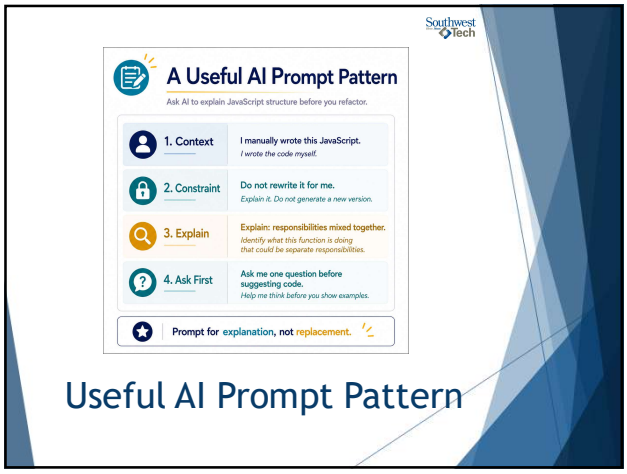
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A Useful AI Prompt Pattern
Ask AI to explain JavaScript structure before you refactor.
1. Context
I manually wrote this JavaScript.
I wrote the code myself.
2. Constraint
Do not rewrite it for me.
Explain it. Do not generate a new version.
3. Explain
Explain: responsibilities mixed together.
Identify what this function is doing that could be separate responsibilities.
4. Ask First
Ask me one question before suggesting code.
Help me think before you show examples.
Prompt for explanation, not replacement.
Useful AI Prompt Pattern

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Demo
"Messy Working Code"

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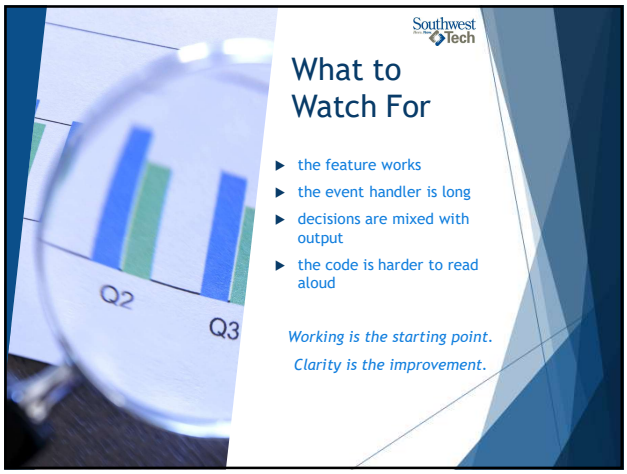

What to Watch For

- the feature works
- the event handler is long
- decisions are mixed with output
- the code is harder to read aloud

Working is the starting point.

Clarity is the improvement.

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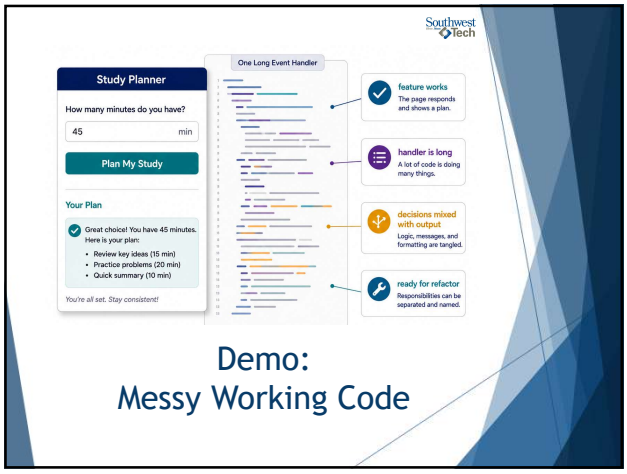

Demo: Messy Working Code

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Find The Responsibilities

Possible responsibilities:

- read the minutes input
- decide which plan fits
- update the message on the page
- respond to the button click

Possible function names:

- `getMinutesAvailable()`
- `chooseStudyPlan(minutes)`
- `showPlan()`

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Hidden Responsibilities → Function Names
Break a long handler into clear, named jobs.
Hidden in One Long Handler
Clear Function Names
read minutes input
Get the number of minutes from the page.
getMinutesAvailable()
Get the number of minutes from the input.
choose study plan
Decide which plan fits the minutes available.
chooseStudyPlan(minutes)
Return the plan that fits the minutes available.
update message
Show the plan and next steps on the page.
showPlan()
Update the message with the plan and next steps.
respond to click
Run everything when the button is clicked.
Name the job before moving the code.
Responsibilities To Functions

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Lab
First refactor pass
Your goal:
choose existing working JavaScript
identify repeated or mixed logic
create at least 2-3 functions
give each function a clear name
retest the behavior

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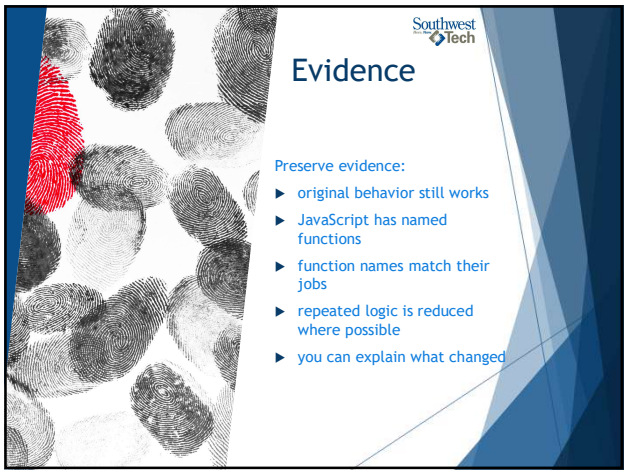
Evidence
Preserve evidence:
original behavior still works
JavaScript has named functions
function names match their jobs
repeated logic is reduced where possible
you can explain what changed

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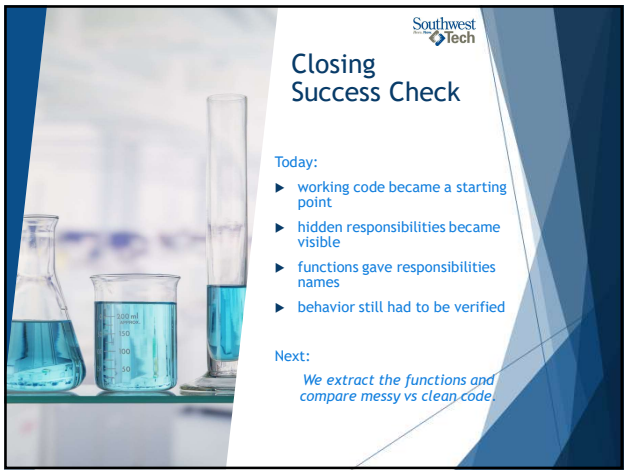
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Closing
Success Check
Today:
working code became a starting point
hidden responsibilities became visible
functions gave responsibilities names
behavior still had to be verified
Next:
We extract the functions and compare messy vs clean code.

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Thank you!
Have Fun!

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